

Estimating the impact of air pollution reduction on premature mortality in Belgium

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- Reducing traffic-related air pollution could substantially lower premature mortality in Belgium
- The g-computation approach to assess potential impact fractions of interventions represents a straightforward approach for drawing causal inferences from observational data

The adverse effects of air pollution on premature mortality are well-documented, yet methodological challenges persist in assessing the health benefits of environmental interventions. This study applies a g-computation approach to estimate the impact of hypothetical air quality interventions on premature mortality in Belgium, using nationally representative survey data.

Methods

- **National Health Interview Surveys data** (BHIS 2008-2013-2018, n = 27,536) linked to environmental data at participants' residential addresses and mortality follow-up through 2023
- **Premature mortality** was defined as death occurring before age 70.
- **Cox proportional hazards models** were used to estimate the association between long-term air pollution exposure and premature mortality, adjusting for socio-economic, environmental, and lifestyle factors
- A **g-computation approach** simulated scenarios where air pollution exposure was reduced to the WHO air quality guideline levels (Fig. 1). Potential impact fractions quantified the **proportion of premature deaths that could be prevented**.

Results

- During follow-up, 3,995 deaths occurred, and 771 participants were lost due to registry deletion.
- Mean follow-up was 8 years. Overall, 2.4% of the study population experienced premature mortality.
- **All-cause premature mortality was significantly associated with NO₂ and BC exposure.** A quadratic term for BC improved model fit, revealing a flattening effect at higher exposures.
- Hazard ratios per interquartile range/1 unit increase were: PM_{2.5}: 1.13 (0.83;1.53)/1.03 (0.96;1.10), PM₁₀: 1.13 (0.93;1.38)/1.02 (0.99;1.06), NO₂: 1.30 (1.01;1.69)/1.03 (1.01;1.04), and BC: 1.45 (1.29;1.75)/1.91 (1.36;2.45). No interactions were found with age or sex.
- **Meeting WHO guideline levels for NO₂ would reduce premature mortality risk by 0.55 percentage points (95% CI: 0.06;1.04), preventing 24% of cases.**
- A **clear dose-response** was observed between NO₂ and BC reductions and premature mortality (Fig 2.)

Fig. 1. Steps of the G-computation approach

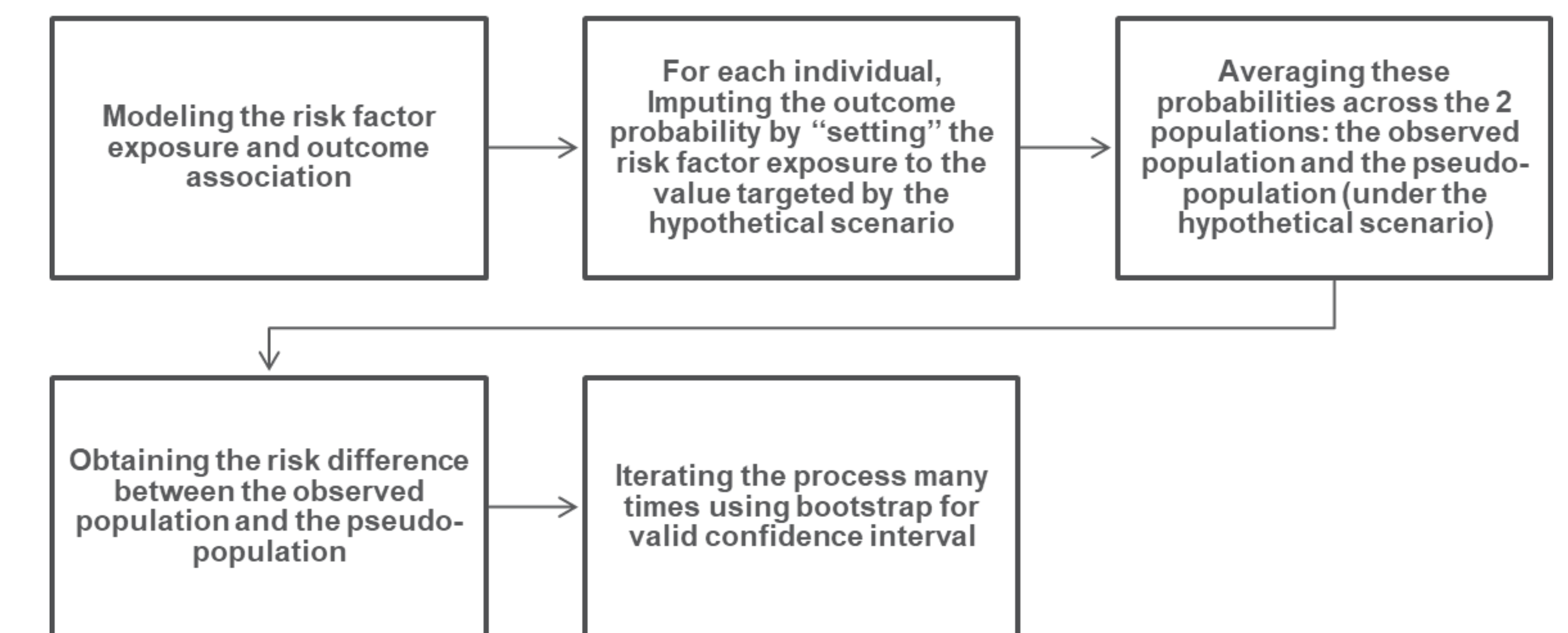


Fig. 2. Dose-response association between air pollution reduction and the potential impact fractions on premature mortality in Belgium. Error bars represent the 95% CI

